CALIBO AI ACADEMY

Customer Churn Prediction using Logistic Regression

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1. Assignment Overview

The objective of this assignment is to use customer data to identify customers who are likely to leave a telecom company. Customer churn is an important business problem because losing an existing customer reduces recurring revenue and usually creates an additional acquisition cost when the company replaces that customer. A predictive model can help the company move from reacting after a customer leaves to taking preventive retention action before the customer churns.

For this task, the Telco Customer Churn dataset is analyzed and a Logistic Regression classification model is used. Logistic Regression is suitable because the target variable has two outcomes: Churn = Yes or Churn = No. The model estimates the probability that a customer will churn and then converts that probability into a final classification.

2. Dataset Understanding

The dataset contains 7,043 customer records and 21 attributes. The target variable is Churn. The available information covers customer demographics, relationship information, phone and internet services, security and support services, contract type, billing preferences, payment method, monthly charges, and total charges.

There are 18 object-type columns, two integer columns and one floating-point column in the original file. The customerID column is an identifier rather than a meaningful predictive feature, so it should not be used as an input to the model. The TotalCharges column is stored as text even though it represents a numeric amount, so it must be converted to numeric before modeling.

3. Data Quality Findings

The initial inspection shows no completely missing values reported by pandas, but there are 11 blank values inside TotalCharges. These blanks occur because TotalCharges is stored as an object column. They need to be converted to NaN during numeric conversion and handled before model training. There are no duplicate rows in the dataset.

The churn distribution is imbalanced but still usable for a baseline classification model: 5,174 customers did not churn (73.46%) and 1,869 customers churned (26.54%). This means accuracy alone should not be used to judge the model because the business is particularly interested in identifying the smaller churn group.

4. Important Business Insights from the Data

Contract type shows one of the clearest differences. Customers on month-to-month contracts have a churn rate of about 42.7%, compared with approximately 11.3% for one-year contracts and only 2.8% for two-year contracts. This suggests that customers without a long-term commitment are a major retention priority.

Tenure also shows a strong pattern. Churned customers have an average tenure of about 18 months, while customers who stay have an average tenure of about 38 months. The median tenure is 10 months for churned customers compared with 38 months for customers who stay. This indicates that newer customers are substantially more vulnerable to churn.

Internet service is another important signal. The churn rate is about 41.9% for Fiber optic customers, compared with 19.0% for DSL and 7.4% for customers with no internet service. Customers without OnlineSecurity or TechSupport also show much higher churn rates than customers who have those services.

Payment method also provides a useful retention signal. Customers using electronic check have a churn rate of about 45.3%, considerably higher than customers using automatic bank transfer, automatic credit card, or mailed check. Customers without partners or dependents also show higher churn rates than customers with those relationships.

Churned customers have higher average MonthlyCharges, approximately 74.44 compared with 61.31 for customers who stay. Average TotalCharges are lower for churned customers because churned customers generally have shorter tenure. This is a useful reminder that individual variables should be interpreted in combination rather than in isolation.

5. Modeling Approach

The workflow begins with data loading and inspection, followed by cleaning TotalCharges and removing customerID from the predictors. The Churn target is converted from Yes/No into 1/0. The data is divided into training and testing sets using a stratified split so that the churn proportion is preserved in both sets.

Categorical variables are converted into numerical form using one-hot encoding. Numeric variables are standardized so that features such as tenure, MonthlyCharges and TotalCharges are placed on comparable scales. These preprocessing steps are combined with Logistic Regression in a single pipeline. This prevents preprocessing leakage and makes the final workflow easier to reproduce.

6. Baseline Model Results

Using an 80:20 stratified train-test split with random_state = 42, the baseline Logistic Regression model achieves approximately 80.5% accuracy and an ROC-AUC of approximately 0.836. Its churn precision is approximately 65.2%, churn recall is approximately 57.5%, and churn F1-score is approximately 61.1%.

The confusion matrix is: True Negatives = 918, False Positives = 115, False Negatives = 159, and True Positives = 215. Therefore, the model correctly identifies 215 customers who actually churned. It incorrectly flags 115 customers who actually stay as churn risks, while it misses 159 customers who actually churn.

7. Business Interpretation

For a retention campaign, the most serious error is usually a False Negative: a customer is actually going to leave, but the model predicts that the customer will stay. This means the company may not provide a retention offer and can lose the customer and future revenue. A False Positive is also costly because the company may spend money offering an unnecessary discount or incentive to a loyal customer, but the customer is still retained.

Therefore, if the company has a limited retention budget, the model should be evaluated with particular attention to recall for the churn class. A business may deliberately lower the probability threshold below 0.50 to identify more potential churners, accepting more false positives in exchange for fewer missed churners.

8. What I Learned from This Assignment

This assignment demonstrates that a data analytics problem is not only about training a machine-learning algorithm. The important steps are understanding the business problem, inspecting data quality, selecting meaningful predictors, preparing mixed numerical and categorical data, training the model without data leakage, evaluating the confusion matrix, and translating model performance into a business decision.

I also learned why accuracy can be misleading in an imbalanced classification problem. A model must be evaluated using precision, recall, F1-score, ROC-AUC and the confusion matrix. Most importantly, the choice of the evaluation metric depends on the business cost of different mistakes.

9. One Improvement for More Reliable Predictions

A practical improvement would be to tune the classification threshold based on the company's retention budget and the relative cost of False Negatives and False Positives. Instead of automatically using 0.50, the company could choose a lower threshold if missing a likely churner is more expensive than contacting an additional loyal customer. A second-stage improvement could be hyperparameter tuning or comparison with tree-based models, but threshold optimization is a simple and directly business-focused improvement for this Logistic Regression solution.

10. Final Conclusion

The Logistic Regression model provides a useful baseline system for identifying customers at risk of churn. The analysis shows that short tenure, month-to-month contracts, fiber-optic service, lack of support/security services, electronic-check payment and higher monthly charges are associated with higher churn levels. The model correctly identifies a meaningful portion of actual churners, but it also misses some customers who leave. The next step for a real telecom company would be to optimize the decision threshold and connect the predictions to a retention strategy so that high-risk customers can receive timely and cost-effective offers.